

Geometry-Aware JEPA for EEG

A controlled frozen-transfer study on TUAB

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Hack the World(s) – EEG Track · Team **HelloWorlds**

Contribution

Our contribution, in one slide

1. **A strong frozen baseline.** SIGReg-ambient JEPA matches the LeJEPA/Laya recipe's value on our (pipeline-helped) setup – not a controlled head-to-head.
2. **Nothing beats it.** Ambient/tangent SIGReg & PEIRA, Graph-JEPA, Fourier-JEPA – none beats ambient SIGReg in balanced accuracy (gaps ≤ 0.013).
3. **Latent geometry is the contribution.** Ours separates normal/abnormal $\sim 2\times$ better than random, visualised with **de Surrel's AIRM Riemannian t-SNE** (the right metric for SPD/EEG covariances). Tangent sharpens the *geometry* (best cov-latent silhouette), not the task.
4. **Rigor over spin.** The limits we report – *decodable* $\neq$ *clustered*, sustained vs transient, AIRM edge within noise. An honest negative beats a fragile positive.

Motivation

The question we actually asked

- **EEG foundation models** are everywhere – but most report *fine-tuned* numbers. What survives when you **freeze** the encoder?
- **Self-supervised JEPA** avoids collapse with an *anti-collapse regulariser*. The EEG covariance lives on a curved **SPD manifold**, not in flat Euclidean space.
- **Our pre-registered hypothesis**: making anti-collapse *geometry-aware* (act in the SPD **tangent**) and *distribution-free* (**PEIRA**) beats a plain ambient baseline.
- **What we ship**: a controlled study with an *honest* answer – a strong frozen probe, a clean negative result with a geometric mechanism, and label/data-efficiency curves.

Data

Data: TUH Abnormal EEG (TUAB)

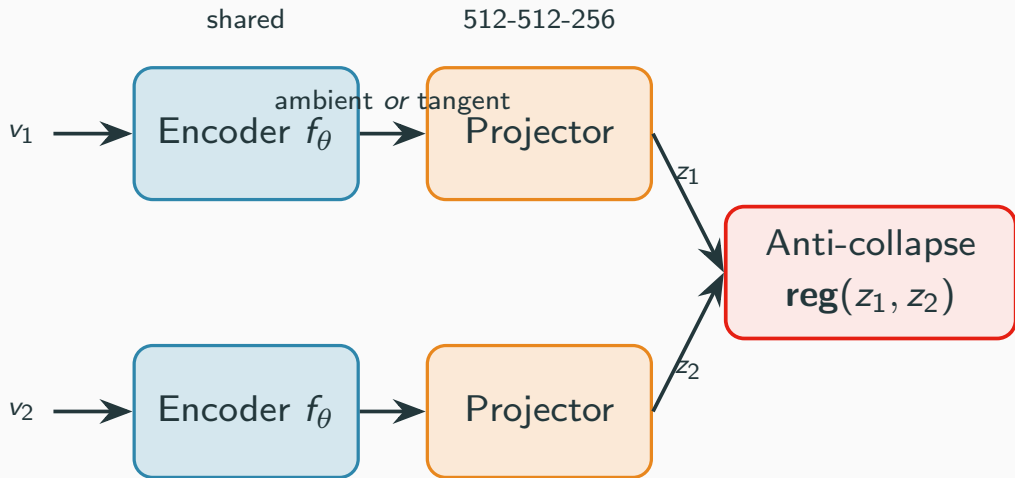
- **Task:** binary *normal* vs *abnormal* EEG, recording level.
- **Signal:** 19 channels @ 200 Hz; we sample 10 s windows (2,000 samples), per-channel z-scored.
- **Standard split:** 2,717 train / 276 eval recordings, **patient-disjoint** (2,076 vs 253 patients, overlap = 0 – verified).
- **Pretraining:** SSL reads only *unlabeled* train recordings; labels enter at the probe alone.

Why we don't chase accuracy

TUAB labels come from clinical reports and are known to be noisy; we are not aware of a published label-noise / Bayes-ceiling estimate. **Sub-1-point gaps without multi-seed CIs are not meaningful** – so we report *balanced accuracy on the full split*, 3 seeds, with error bars.

Architecture

Two view and 1D convolutional encoder



What we tried

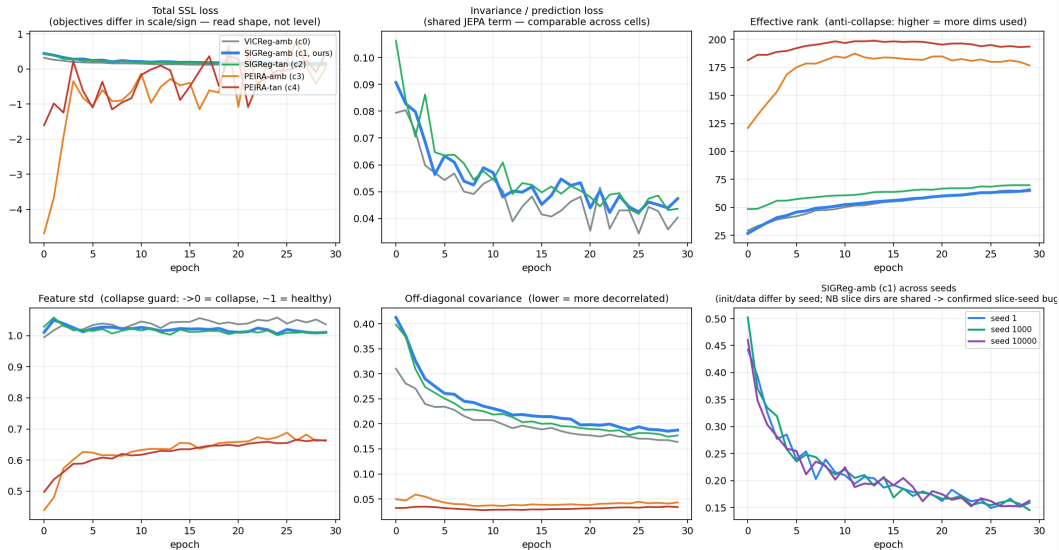
regulariser \ space	ambient (Euclidean)	tangent (SPD)
VICReg (variance/cov)	C0 reference	—
SIGReg (Gaussianity slices)	C1 <i>Laya-like baseline</i>	C2 geometry-aware
PEIRA (distribution-free)	C3	C4 <i>ex-hypothesis</i>

- **SIGReg** (LeJEPa/BCS): tests random 1-D slices for Gaussianity – assumes an *isotropic* target.
- **PEIRA**: distribution-free anti-collapse – no target distribution to mis-specify.
- **Hypothesis**: the geometry-aware / distribution-free cells (C2, C4) beat the ambient baseline C1.
- **GraphJEPa**
- **FourierJEPa**

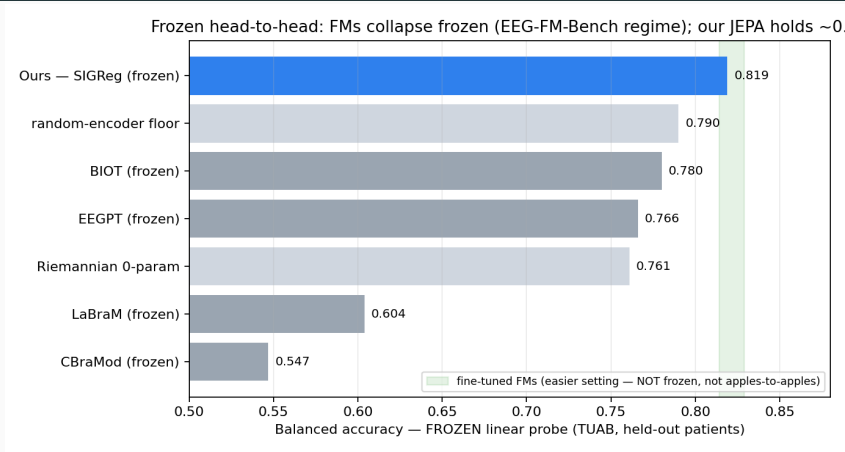
Results

SSL training curves (all cells healthy)

EEG-JEPA SSL training curves — 5 regularizer cells, 30 epochs, TUAB pretrain (seed 1)



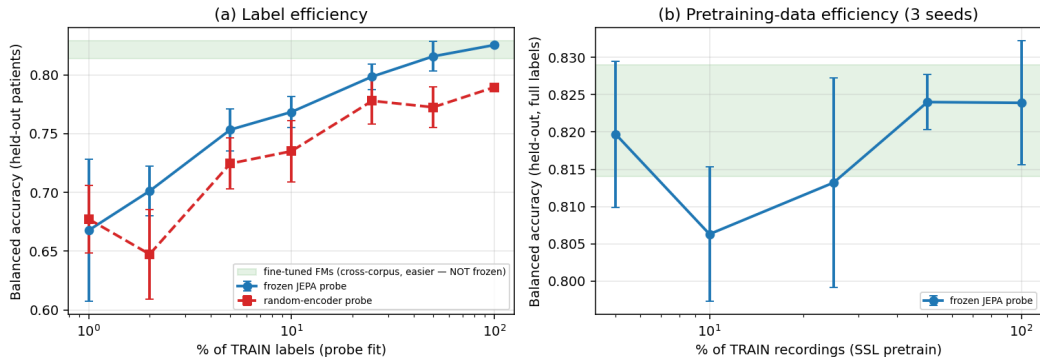
Frozen & in-domain: we hold where frozen FMs fall



Frozen linear probe, TUAB held-out patients. **Ours 0.819** (3-seed) sits above the random floor and every frozen FM baseline; fine-tuned FMs (green band) are an *easier*, non-comparable setting.

The value of self-supervision (label & data efficiency)

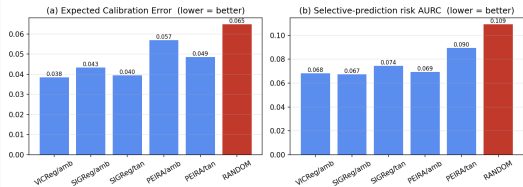
Value of self-supervision — frozen in-domain EEG-JEPA on TUAB



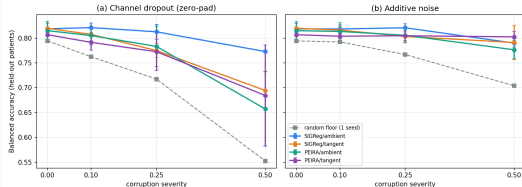
(a) The money plot: *JEPA @ 25% labels* $\approx$ *random-encoder @ 100%* — $\sim 4\times$ label efficiency, $\sim +0.04$ BalAcc. (b) Probe is stable down to small pretraining-data fractions.

Beyond accuracy: calibration & robustness (3 axes, not 1)

Frozen-probe reliability on TUAB — SSL beats the random floor; geometry/PEIRA does not beat ambient (1 seed)



Frozen-probe robustness on TUAB (3-seed mean $\pm$ std) — ambient SIGReg is the most dropout-robust

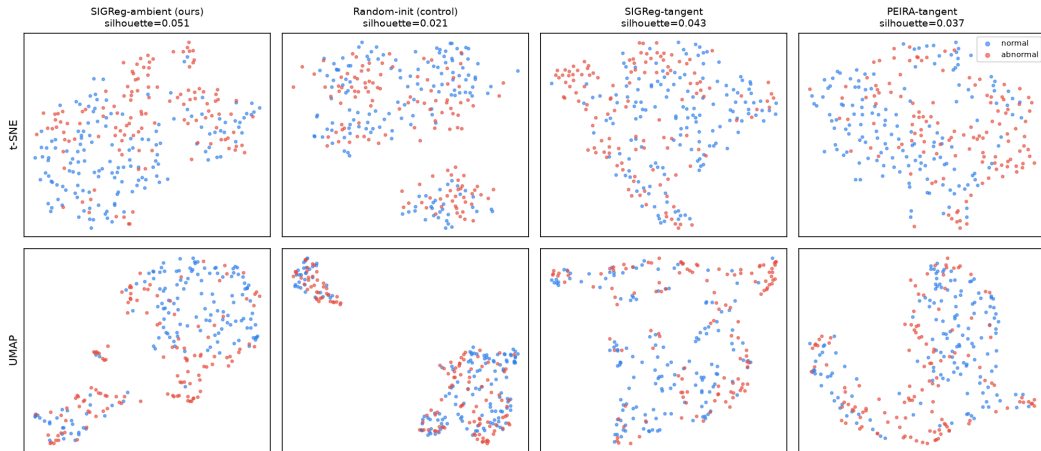


- **Calibration:** every SSL encoder beats the random floor (ECE, selective prediction) – geometry/PEIRA does *not* beat ambient.
- **Robustness (3-seed):** ambient SIGReg is the *most* channel-dropout-robust; the SPD-tangent objective *degrades* dropout-robustness; no geometry win under noise (the single-seed hint did not replicate).
- **The null holds on all three axes** – accuracy, calibration, robustness. Geometry buys none of them here.

Clustering

Frozen latent space (ours vs random vs geometry)

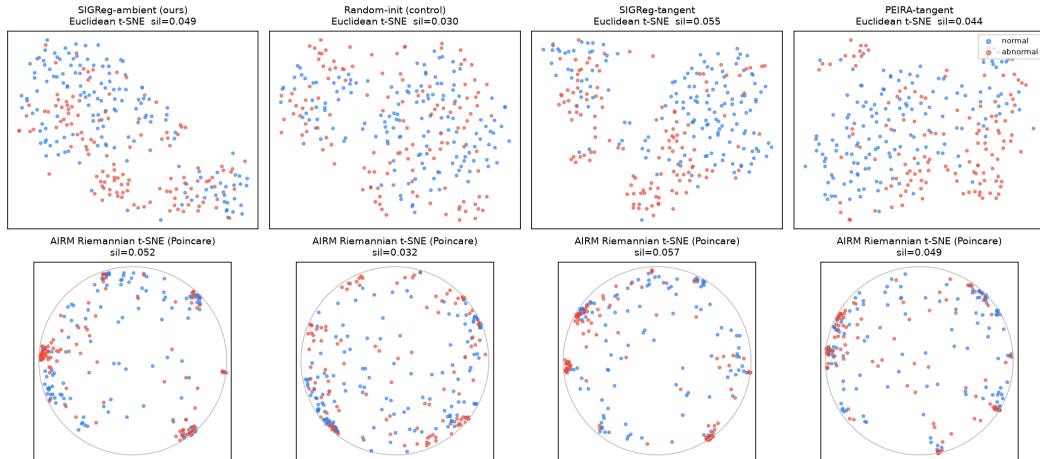
Frozen latent space on TUAB eval — normal vs abnormal (recording-level)
same frozen ambient represent() readout for all; silhouette is on full-dim features (the 2-D map is illustrative)



t-SNE and UMAP of frozen represent features on the eval set – same frozen ambient readout for all

Manifold-aware latent geometry (AIRM vs Euclidean)

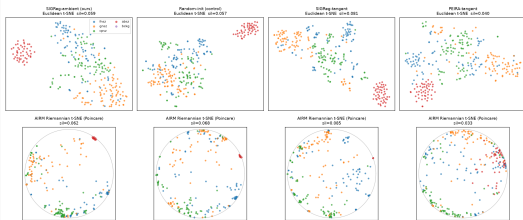
Frozen SPD latents on TUAB eval — Euclidean (tangent) t-SNE vs AIRM Riemannian t-SNE
silhouette is full-dim (Euclidean tangent vs AIRM geodesic); 2-D maps are illustrative



Manifold viz across TUH tasks – sustained vs transient

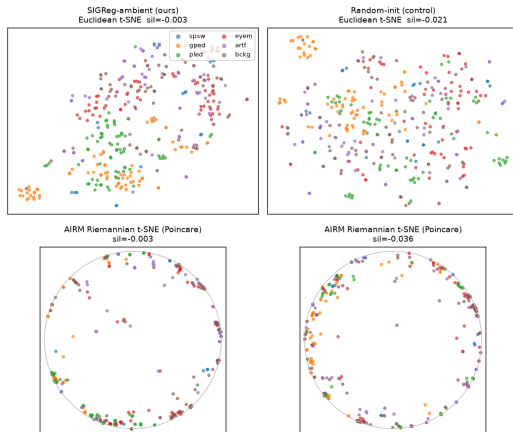
TUSZ – 5 classes; SIGReg-tangent separates best

Frozen SPD latents on TUSZ eval – 5 seizure types (cross-task, same-site)
Euclidean (tangent) t-SNE vs AIRM Riemannian t-SNE; silhouette is f1-f2-dm



TUEV – 6 event types (decodable, not clustered)

Frozen SPD latents on TUEV eval – 6 event classes (cross-task, same-site)
Euclidean (tangent) t-SNE vs AIRM Riemannian t-SNE; silhouette is full-dim



DECODABLE (frozen probe BA ~0.40, >> chance 0.17; ours > random) but NOT CLUSTERED (silhouette ~0)
transient sub-second events diluted by the 10 s window -- silhouette measures clustering, not decodability

task (#cls)

sil_{Euc}

sil_{AIRM}

- Clusters **sustained** phenomena (abnormality, seizures), not

Caveats we disclose to the jury

- **Report 0.819, not 0.833.** The single-seed 0.833 (SIGReg-ambient, seed 1) was the *top* of that cell's range; the 3-seed mean is 0.819.
- **SIGReg slice-seed bug (found, fixed).** BCS seeded slice directions per-step, so the 3 SIGReg seeds shared one slice schedule – error bars *understate* variance. Now fixed (run-seeded RNG); the reported numbers predate the fix, not re-run.
- **PEIRA tangent (0.807) is not evidence against wrapped-Gaussian.** PEIRA is distribution-free $\Rightarrow$ immune to the isotropic-target mis-specification $\Rightarrow$ it never tests that hypothesis. 0.807 vs 0.815 is within noise.
- **Bootstrap CI (illustrative, not yet run).** A bootstrap over the 276 eval recordings would plausibly give a ± 0.02 – 0.03 CI – wider than the max inter-cell gap (0.013) – supporting “benchmark-resolution-limited”. The resampling is not yet committed.
- **Preprocessing differs** from the published FM rows – the head-to-head is indicative, not a controlled re-run of their pipelines.